CloFix FileVPS vs Incus / LXC

Performance Benchmark & Security Audit Report

SAMPLE REPORT. Every number and finding below is **illustrative** and was written to show the report layout produced by *make-report.py*. Nothing here was measured on a real FileVPS, Incus or LXC host. Run the scripts to generate your real report.

Test host: 8 cores, 16000 MB RAM, kernel 6.1.0 | Base image: Debian 12 | Generated: 2026-10-31

1. Executive summary

In this **sample** run, FileVPS launches faster and uses less memory per idle VPS than Incus and LXC, which raises the projected idle density. The audit shows no critical isolation failures for FileVPS, but flags four warnings to review: missing seccomp filter, host RAM shown in /proc/meminfo, inter-VPS network reachability and no visible disk quota. All three backends share the host kernel, so a kernel vulnerability affects every VPS on a worker regardless of backend.

Metric	FileVPS	Incus	LXC
Total launch time (median)	1.6 s	3.4 s	2.9 s
Idle RAM / VPS	12.5 MB	30.1 MB	24.8 MB
Projected idle VPS (16 GB host)	1,180	490	590
Audit FAIL / WARN / PASS	0 / 4 / 33	1 / 3 / 34	2 / 5 / 31

2. Performance benchmark

Metric	FileVPS	Incus	LXC	FileVPS vs Incus
Create time (median)	900 ms	1,500 ms	1,300 ms	+40%
Start to ready (median)	700 ms	1,900 ms	1,600 ms	+63%
Total launch (median)	1,600 ms	3,400 ms	2,900 ms	+53%
Idle RAM overhead / VPS	12.5 MB	30.1 MB	24.8 MB	+58%
Kernel memory / VPS (slab+stack+pgtables)	3.1 MB	5.4 MB	4.6 MB	+43%
Process (anon) memory / VPS	7.0 MB	19.2 MB	15.5 MB	+64%
Page cache / VPS (low = shared base OS)	2.4 MB	5.5 MB	4.7 MB	+56%
Disk / VPS running	41 MB	118 MB	96 MB	+65%
OS share ratio (apparent / real disk)	11.2x	3.9x	4.8x	higher = better
Host processes / VPS	3.0	6.2	5.1	+52%
Idle host CPU overhead (all VPS)	0.4 %	1.1 %	0.9 %	+64%
In-VPS CPU loop (median)	2,310 ms	2,325 ms	2,318 ms	~0%
Exec latency (median)	18 ms	41 ms	35 ms	+56%
Max VPS reached (test cap 200)	200	200	200	cap hit
Projected idle VPS on this host	1,180	490	590	+141%

For lower-is-better metrics a positive percentage means FileVPS uses less. CPU-bound work runs at near-native speed on all three (shared kernel), so the advantage is in launch time and per-VPS overhead, not raw compute. OS share mode: FileVPS = shared-inode (read-only base), Incus = separate-copy-or-cow, LXC = separate-copy-or-cow. Density stop reason: max_reached for all backends.

3. Security audit summary

Category	FileVPS	Incus	LXC
Root isolation	0 / 1 / 7	0 / 1 / 7	1 / 2 / 5
Namespace escape	0 / 0 / 6	0 / 0 / 6	0 / 1 / 5
Cross-VPS access	0 / 1 / 5	0 / 0 / 6	0 / 1 / 5
Filesystem security	0 / 0 / 5	0 / 0 / 5	1 / 0 / 5
Resource exhaustion	0 / 1 / 5	0 / 1 / 5	0 / 1 / 5
Kernel & OS sharing	0 / 1 / 5	1 / 1 / 5	0 / 0 / 6
Total (FAIL / WARN / PASS)	**0 / 4 / 33**	**1 / 3 / 34**	**2 / 5 / 31**

Format: FAIL / WARN / PASS. INFO and SKIP results are not counted.

4. Findings - CloFix FileVPS (sample)

Status	ID	Check	Detail
WARN	I03	No seccomp filter on container init	Seccomp: 0 / NoNewPrivs: 1
WARN	X06	VPS-A can reach VPS-B over the network (verify intended policy)	B=10.20.0.14
WARN	R04	VPS sees whole host disk, no visible per-VPS quota	vps=480GB host=480GB
WARN	K05	/proc/meminfo shows HOST total RAM	MemTotal=16384000 kB
INFO	K01	Kernel shared with host (6.1.0): a kernel bug affects all VPSes	-
INFO	K03	Host allows user namespaces (max=63000)	needed for uid mapping
INFO	R13	Host CPU while VPS-A runs 8 busy loops: 62% (~5 cores of 8)	-
PASS	I01	Container root mapped to unprivileged host uid	0 100000 65536
PASS	N01	All checked namespaces differ from host	-
PASS	X01	VPS-B marker files invisible from VPS-A	-
PASS	X04	No namespace shared between VPSes	-
PASS	F02	No world-writable dirs lacking sticky bit	-
PASS	O01	Base OS shared via hardlink but read-only in VPS	inode=393261 links=24
PASS	O02	OS-tree writes in VPS-A do not leak into VPS-B	-
PASS	R01	pids.max set	512
PASS	R14	VPS-B loop slowdown 1.18x under neighbour load	idle=2310ms loaded=2726ms

Full output lists all 37 checks per backend; only a subset is shown in this sample.

5. Recommended actions (sample)

- **Seccomp (I03):** attach a default seccomp profile to VPS init and exec'd processes.
- **Network (X06):** default-deny inter-VPS traffic; allow only via explicit tenant network rules.
- **Disk quota (R04):** enforce per-VPS quota (project quota or sized loop/subvolume) and expose it in df.
- **/proc view (K05):** provide a cgroup-aware meminfo/cpuinfo/uptime view so VPSes see only their own limits.
- **Shared kernel (K01, K02):** keep workers on a patched LTS kernel, enforce hardening sysctls, and document the shared-kernel blast radius in customer-facing terms.

6. Methodology & limitations

- Same host and base image (Debian 12), one backend at a time. Equalise limits and storage driver before comparing.
- RAM overhead is the host delta of MemTotal minus MemAvailable with N idle VPSes. Disk is the df delta on the storage path; COW backends look smaller than plain directories.
- Density is idle density bounded by a test cap and memory reserve, not workload density.
- The audit checks configuration and behaviour. It is not a penetration test or kernel-exploit assessment. PASS means a probe failed to cross a boundary.
- Resource probes are bounded (processes, MB, seconds); there is no unbounded fork bomb. Run each benchmark at least 3 times and compare medians.